

AI Video Generation Architectural Audit

Current-state workflow, competitive benchmark and target architecture

Scope

User input through final render and export

Assessment

18 production stages, dependencies, bottlenecks, duplication, performance, scalability and missing capabilities

PREPARED 27 JULY 2026

Read-only audit - no source code modified

AUDIT CONCLUSION

01. Executive assessment

The current product is a deterministic script-to-ad assembler rather than a complete AI creative platform. It has valuable production foundations - ElevenLabs timing, deterministic variation, licensed-media providers, Plandome branding, Hyperframes rendering, FFmpeg validation and offline fallbacks - but it lacks a shared creative project model connecting story, visual intent, assets, style, timeline, editor and renderer.

PRIMARY DIAGNOSIS

The quality issue is not a shortage of effects. It is a coordination problem. Several independent planning and styling systems make local decisions without one canonical creative brief or timeline. The renderer can therefore produce a technically valid video whose imagery is irrelevant, whose style is internally inconsistent, or whose template treatment is inappropriate for the scene.

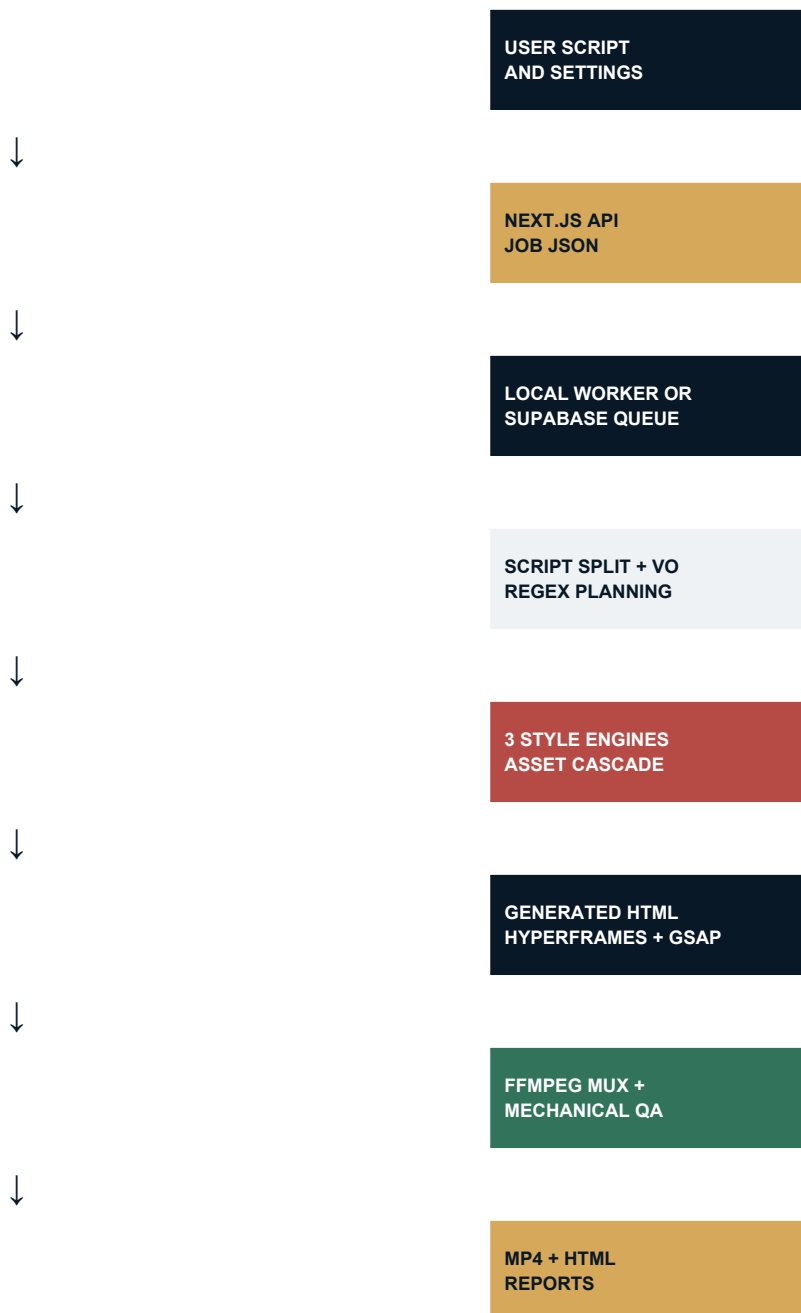
- Relevant visuals: keyword matching and fallback generation can pass mechanical validation without semantic relevance.
- Remotion: the active production path uses generated Hyperframes HTML and GSAP. Remotion is not an installed runtime dependency.
- Creative direction: three overlapping style systems may contradict one another.
- Editor: visible scene edits are stored, but only a small preference subset influences later rendering.
- Captions: word-level clips inflate the timeline and can force transitions to be disabled.
- Operations: file/object-storage job state and queue semantics are not durable enough for production scale.

RECOMMENDED ARCHITECTURAL TARGET

One versioned CreativeProject, one art-direction compiler, one multi-track timeline, a typed template registry, semantic asset intelligence, a non-destructive editor and a durable render workflow.

RUNTIME TOPOLOGY

02. Current architecture



The main critical path is concentrated in scripts/video-worker.ts. It coordinates narration, planning, media resolution, style selection, composition generation, rendering, muxing, quality checks and artifact publication. This creates high coupling and makes partial retries difficult.

CURRENT STATE, FLOW AND RISKS

03. The 18-stage workflow

Each row records what currently happens, the information entering and leaving the stage, and the most important architectural limitation. Dependencies, performance and scaling effects are expanded in the following sections.

#	Stage	Input	Output	Current limitations
1	User input	Raw script, render mode, avatar flag, optional remote media.	Validated job JSON.	Thin Zod contract; duplicated UI/backend splitting; no audience, objective, claims, platform, duration, brand brief, moderation or ownership.
2	Prompt engineering	Scene text, keyword rules, provider configuration.	Search strings and provider-specific prompts.	No central prompt compiler, versions, evaluations, negative constraints or provenance. Query logic is scattered.
3	Story generation	Submitted script.	Unchanged narration.	No hook generation, story arc, benefit sequencing, claim review, duration rewriting or concept alternatives.
4	Storyboard	Sentence-like script fragments.	Scene array; post-render HTML storyboard.	Not a pre-production approval artifact. No thumbnails, shot references, locks, regeneration or continuity notes.
5	Scene planning	Narration fragments and approximate timing.	Kinds, headlines, visual briefs and queries.	Regex-driven; multiple overlapping planners; no canonical scene schema; no narrative or shot continuity.
6	Asset selection	Queries, scene category, aspect ratio.	Downloaded or generated image/video plus attribution.	User media, Drive, cache, Pixabay, Commons, ComfyUI, fallback. Weak semantic ranking; duplicated provider code; no indexed manifest.

03. The 18-stage workflow - continued

#	Stage	Input	Output	Current limitations
7	Style selection	Seed, history and scene count.	Palette, fonts, layouts, camera and transitions.	Creative System, Visual Variety and Design DNA overlap. Randomness outweighs semantic fit; process-local history is ineffective.
8	Template selection	Style metadata and scene index.	Feature/layout names.	Local arrays and CSS adaptations, not actual Remotion templates. Features are assigned regardless of scene relevance.
9	Motion planning	Scene type, duration, template feature.	CSS classes and GSAP calls.	No motion-plan IR; intent is coupled to implementation; captions can suppress transitions.
10	Camera planning	Seeded scene metadata.	Pan, zoom and parallax transforms.	No shot size, angle, screen direction, subject tracking or first/last-frame continuity.
11	Timeline	Narration alignment or text-length weights.	Scene timings and per-word clips.	No canonical multi-track timeline. One object per word causes DOM and render growth; editor timing is not authoritative.
12	Components	Design recipes and feature labels.	HTML fragments and classes.	Weakly typed; many recipes are metadata only; no manifests, capability contracts, previews or component regression tests.

03. The 18-stage workflow - continued

#	Stage	Input	Output	Current limitations
13	Composition	Scenes, assets, styles, captions and logo.	Large generated HTML document.	No active Remotion composition. Output strings are difficult to maintain and do not represent an editable project.
14	Animation	Feature classes and timings.	GSAP/CSS browser timeline.	String-generated animation; multiple libraries; no typed keyframes or easing tokens; poor component isolation.
15	Audio	Script and provider credentials.	Narration, duration and character alignment.	ElevenLabs is useful, but music, SFX, ducking, mastering, pronunciation, stems and localisation are missing.
16	Subtitles	Script and word alignment.	Animated burned-in captions.	Overly granular word treatment; no SRT/VTT, transcript editor, reading-speed checks, OCR safety or multilingual tracks.
17	Rendering	HTML composition, assets and narration.	H.264/AAC MP4 and QA reports.	Detached processes and object-storage queue; no lease, checkpoint, cancellation, robust retry or distributed scene rendering.
18	Export	Rendered media and metadata.	MP4, inspector, attribution and HTML storyboard.	No range streaming, variants, editable interchange, native Canva project, captions/stems, publishing or version history.

CROSS-CUTTING FINDINGS

04. Dependencies and operational risks

The pipeline depends on four distinct execution layers:

- Application: Next.js, React, API routes and Zod contracts.
- Creative/runtime: local TypeScript planners, HTML composition, GSAP, SplitType, Motion and Hyperframes.
- Media: ElevenLabs, HeyGen, Pixabay, Wikimedia Commons, Google Drive assets, optional ComfyUI and FFmpeg.
- Persistence: local JSON/filesystem or Supabase Storage objects.

The most consequential operational risks are:

- Job loss: the remote daemon removes a queue object before work has completed.
- No idempotency: a retry can repeat expensive provider and rendering work.
- No checkpointing: late failures can force an almost complete rerun.
- No multi-tenancy boundary: job routes do not establish robust user/workspace ownership.
- No indexed asset cache: reuse and history rely on scanning directories and JSON files.
- No provider control plane: rate limits, costs, latency and health are not centrally managed.

Duplicate logic

Area	Current duplicates	Recommended owner
Creative direction	Creative System, Visual Variety, Design DNA	ArtDirectionEngine
Scene intelligence	Script scenes, universal planner, premium planners	StoryboardPlanner
Asset retrieval	Provider modules plus legacy direct Commons searches	AssetProvider adapters
Composition	Premium, legacy and scene-renderer variants	Renderer registry
Persistence	Local JSON, Storage objects, unused DB migration	Project and Job repositories

CRITICAL PATH

05. Performance and scalability

Sequential media work

Current: Search, download and image-to-video conversion sit in the job critical path.

Target: Parallel bounded asset tasks, provider rate limits and scene caches.

Large inline composition

Current: One generated HTML file embeds extensive CSS, markup and timelines.

Target: Typed components and a renderer-independent project/timeline model.

Caption density

Current: Timeline and DOM complexity grow with every spoken word.

Target: Phrase-level caption clips with active-word spans.

Repeated rendering

Current: A late failure reruns expensive completed stages.

Target: Stage checkpoints and content-addressed outputs.

Filesystem scans

Current: Audio reuse and creative history scale with total job count.

Target: Indexed metadata and content hashes.

Single-job worker model

Current: Detached local processes and a simple daemon provide weak back-pressure.

Target: Durable workflow engine, leases, concurrency pools and cancellation.

MODERN AI CREATIVE PLATFORMS

06. Competitive benchmark

Capability	Current system	Market direction
Idea-to-script	Missing	InVideo generates scripts, visuals, voiceover, subtitles and music from an idea.
Semantic auto-edit	Keyword heuristics	Canva assembles footage with smart cuts, synced audio, transitions and story-aware layouts.
Conversational editing	Missing	Captions and Adobe support natural-language changes and focused edits.
Editable design canvas	Basic scene form	Canva and Adobe expose layers, assets, captions and manual adjustment.
Camera control	Pan/zoom presets	Adobe, Veo, Pika and generative-video systems expose shot and camera controls.
Reference consistency	Missing	Runway, Veo and Kling use references for characters, products, objects and locations.
Multi-shot continuity	Missing	Kling and Veo provide multi-shot or first/last-frame continuity controls.
Audio direction	Narration only	InVideo and Captions add music, SFX, cleanup and automatic mixing.
Localisation	Missing	Captions provides translation, dubbing, voice continuity and lip sync.
Template ecosystem	Local arrays/CSS	Canva, Adobe and professional editors expose real editable components and templates.

STRATEGIC INTERPRETATION

The product does not need to recreate every frontier video model. It needs a provider-neutral creative architecture capable of routing selected shots to stock, internal assets, ComfyUI, Veo, Runway, Kling or Pika while retaining narrative, brand, timeline and editorial control.

CAPABILITY GAPS

07. Missing quality subsystems

Canonical CreativeProject

Versioned brief, story, scenes, assets, design tokens, timeline, audio, captions, provenance and exports.

Story and script engine

Hooks, arcs, alternatives, claim review, pacing and approval.

Multimodal asset intelligence

Embeddings, VLM reranking, duplicate detection, product/logo recognition and licence tracking.

Provider router

Capability, quality, price, latency, quota and safety-aware model selection.

Canonical timeline

Video, VO, music, SFX, captions, overlays, keyframes and transitions.

Non-destructive editor

Canvas, layers, tracks, trim, split, keyframes, asset replacement, history and prompt edits.

Multimodal quality evaluator

OCR, relevance, safe zones, product prominence, brand compliance, continuity and aesthetics.

Creative learning

Project revisions, explicit preferences, approvals, A/B variants and campaign-performance feedback.

Brief interpreter

Audience, objective, offer, tone, platform, duration, claims, brand constraints and CTA.

Semantic storyboard

Shot planning, references, continuity, lock/regenerate and pre-render approval.

Art-direction compiler

One design language compiled into tokens, components, motion, camera, captions and transitions.

Continuity manager

Products, characters, locations, palette, wardrobe, geometry and first/last frames.

Template registry

Versioned components with manifests, previews, constraints and asset requirements.

Audio director

VO, music, SFX, ducking, mastering, pronunciation, stems and localisation.

Durable orchestration

Retries, leases, checkpoints, cancellation, fan-out, observability and cost.

TARGET STATE

08. Ideal architecture



DELIVERY ROADMAP

09. Prioritised improvements

HIGH - ARCHITECTURAL FOUNDATIONS

- Introduce the versioned CreativeProject intermediate representation.
- Unify all style systems under one art-direction compiler.
- Make editor operations mutate the canonical project and drive regeneration.
- Add multimodal semantic asset ranking and visual relevance QA.
- Implement a durable queue with leases, retries, idempotency and checkpoints.
- Split video-worker.ts into separately testable workflow stages.
- Create a canonical multi-track timeline.
- Move from per-word clips to phrase captions with active-word spans.
- Add music, SFX, ducking, loudness and audio-quality control.
- Add authentication, ownership and project access control.
- Integrate real Remotion components or correct the product description.

MEDIUM - CREATIVE CAPABILITY

- Story and hook generation with approval.
- Storyboard review, scene locking and targeted regeneration.
- Reference-conditioned media and cross-shot continuity.
- Template manifests, capability rules and preview catalogue.
- Scene-level cache, resumable rendering and aspect-ratio fan-out.
- SRT/VTT, localisation, dubbing and conversational editing.
- Golden-render and visual-regression testing.
- Asset provenance, licensing and provider cost routing.

LOW - EXPANSION AFTER FOUNDATIONS

- Additional decorative systems and transition packs.
- More fallback illustration variants.
- Advanced collaboration, comments and approvals.
- Publishing connectors and campaign-performance optimisation.

BOUNDARIES

10. Separation, consolidation and modularity

SEPARATE	MERGE
• API request handling / workflow orchestration • Story generation / scene planning • Scene planning / asset resolution • Asset search / download and transcoding • Style selection / style rendering • Motion intent / renderer implementation • Timeline compilation / composition rendering • Narration generation / audio mixing • Quality assessment / rendering • Export creation / workflow execution • Persistence interface / local or remote backend • Editor UI / project mutation commands	• Style engines -> ArtDirectionEngine • Scene planners -> StoryboardPlanner • Stock searches -> AssetProvider adapters • Renderer variants -> Renderer registry • UI/backend segmentation -> ScriptSegmenter • Job stores -> JobRepository • Design recipes + components -> Template registry • Feedback + scene edits -> Project revisions • History + learning -> CreativeMemory service

MODULAR TARGET

11. Suggested project structure

```
apps/  
  studio/  
    app/ editor/ generator/  
packages/  
  domain/  
    creative-project/ brief/ story/ storyboard/  
    timeline/ assets/ audio/ captions/ exports/  
  creative-engine/  
    brief-interpreter/ script-agent/ storyboard-planner/  
    art-direction/ motion-planner/ camera-planner/  
    quality-evaluator/  
  templates/  
    registry/ editorial/ paper/ tech/ documentary/  
    product-showcase/  
  renderers/  
    core/ remotion/ hyperframes/ ffmpeg/ preview/  
  providers/  
    stock/ generative/ speech/ avatar/  
  orchestration/  
    jobs/ workflows/ workers/ retries/ events/  
  persistence/  
    projects/ jobs/ assets/ provenance/ revisions/  
  observability/  
    logging/ tracing/ metrics/ cost/  
  test-kits/  
    golden-projects/ visual-regression/  
    provider-contracts/ render-fixtures/
```

DESIGN PRINCIPLES

12. Future-proof architecture

- Treat the CreativeProject - not generated HTML - as the source of truth.
- Keep rendering engines replaceable behind a common timeline contract.
- Represent every provider as an adapter with capabilities, cost, latency and safety limits.
- Store prompts, model versions, seeds, references, asset licences and transformations as provenance.
- Version templates independently from saved projects.
- Use scene-level content hashes so unchanged shots are never regenerated.
- Separate low-latency proxy previews from final rendering.
- Make quality failures trigger targeted repairs instead of whole-job restarts.
- Use event-driven workflows with durable checkpoints and idempotent stages.
- Create golden scripts and visual-reference tests before adding template volume.
- Learn from explicit edits and approved versions, scoped by user, brand and campaign.
- Compile multiple aspect ratios and languages from the same canonical timeline.
- Maintain a capability matrix so the UI only promises features the active runtime truly supports.

FINAL RECOMMENDATION

Build toward one canonical creative project, one art-direction system, one timeline, real editable components, semantic assets and a durable render workflow. Additional effects should follow these foundations, not precede them.

REFERENCE

13. Sources and implementation evidence

Repository implementation

scripts/video-worker.ts; scripts/premium-visual-composition.ts; scripts/premium-visual-orchestrator.ts;
scripts/premium-visual-variety.ts; scripts/design-dna.ts; apps/web/lib/video/creative-system.ts;
apps/web/app/(app)/video-editor/video-editor.tsx.

Canva

<https://www.canva.com/newsroom/news/canva-video/>

Adobe Express

<https://helpx.adobe.com/express/web/video-creation-and-editing/create-videos/generate-videos.html>

Adobe AI Assistant

<https://helpx.adobe.com/express/web/ai-assistant/adobe-express-ai-assistant-overview.html>

InVideo AI

<https://invideo.io/make/ai-video-generator/>

Captions AI Edit

<https://captions.ai/features/edit-with-ai>

Captions dubbing

<https://captions.ai/features/ai-dubbing>

Runway Gen-4

<https://runwayml.com/research/introducing-runway-gen-4>

Google Veo

<https://deepmind.google/models/veo/>

Pika

<https://pika.art/faq>

Kling

<https://app.klingai.com/cn/quickstart/klingai-video-3-model-user-guide>

Assessment scope: checked-in repository implementation and publicly documented platform capabilities available at the time of review. Product comparisons describe architectural capabilities, not a claim of feature-for-feature equivalence.